Abba Otieno Ndomo
Atlanta, GA | 404-819-5945 | abba.ndomo@gmail.com | Portfolio: https://portfolio-site-ea5i.onrender.com
Work Authorization: F-1 (CPT eligible) | OPT (May 2027) | STEM OPT eligible | No sponsorship needed

EDUCATION

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Industrial & Systems Engineering – Operations Research *Aug. 2023 – May 2027*

- GPA: 3.5 | Dean's List (4x), Peer Inspiration Award, 3rd Place OKSEF International Engineering Fair
- Coursework: Data Structures & Algorithms, Database Systems, Machine Learning, Stochastic Processes, Linear & Non-Linear Optimization

PROFESSIONAL EXPERIENCE

Quantitative Analyst @ GT Investment Committee Jan. 2025 – May 2025
Python, Bloomberg Terminal API, Pandas, Statsmodels, NumPy *Atlanta, GA*

- Built Python equity screener automating fundamental analysis (P/E, EV/EBITDA, DCF) and technical indicators (RSI, MACD); increased ticker coverage from 50 to 120 stocks
- Automated Bloomberg Terminal API data ingestion using blpapi library, processing real-time market feeds and historical data for $500K portfolio
- Validated sector-rotation strategies using ARIMA time series models and rolling correlation analysis on interest rate data, reducing portfolio VaR by 15%
- Conducted Monte Carlo simulations (10,000 iterations) stress-testing portfolio under various macroeconomic scenarios (rate hikes, recession, inflation shocks)

Data Science Intern Jun. 2024 – Aug. 2024
Python, LangChain, OpenAI API, PostgreSQL, PyTest, Docker *Mumbai, India*

- Engineered RAG-based NLP pipeline using LangChain and GPT-4 to automate high-latency manual audits, achieving 95% precision; reduced field team verification time by 60%
- Built ETL architecture using Pandas and custom PDF parsers (PyPDF2, Tabula) merging unstructured reports with structured SQL databases, processing 10,000+ documents monthly
- Designed PyTest suite with 99% coverage validating model performance across 5 document format categories; prevented 12+ critical bugs from production deployment
- Implemented document classification system routing PDFs to specialized parsers (OCR for scanned, table-detection for digital), improving processing success from 60% to 95%

PROJECTS

Founder @ PerChance Jan. 2026 – Present
Django, Postgres, Docker, ETL, XGBoost

- Engineered fault-tolerant ETL pipeline ingesting 500,000+ NBA records; derived 6 custom SQL features (player efficiency, team momentum, opponent defensive rating, rest days) using Holt-Winters' exponential smoothing to model time-decay
- Trained XGBoost classifier achieving 70% accuracy across 20 analyzed metrics; implemented automated retraining pipeline reducing ML prep time by 90% through Pandas vectorization
- Architected scalable 3-tier Dockerized infrastructure (Django/Postgres/React) serving 100+ daily users; REST API delivers predictions with <200ms p95 latency via Redis caching

Hospital Operations Database Oct. 2025 – Dec. 2025
SQL (PostgreSQL), Python

- Designed PostgreSQL schema with 3NF normalization across 15 tables modeling patient admissions, bed occupancy, staff scheduling; processed 50,000+ patient records
- Executed end-to-end ETL pipeline using Python and Pandas automating data cleaning for 8 disparate CSV sources (20GB total) with null handling, constraint validation, and deduplication
- Developed complex SQL queries with Window functions (LAG, LEAD) and CTEs analyzing bed utilization; created composite indexes on (department_id, admission_date) reducing query time from 3s to 200ms
- Identified 20% latency in discharge-to-admission cycles through timestamp analysis, revealing operational bottlenecks in bed turnover process

Loan Default Risk Model Oct. 2025 – Dec. 2025
SQL, XGBoost, SHAP, Python

- Built XGBoost ensemble classifier on dataset of 50,000+ loan applications achieving 80% AUC and 82% recall; applied SMOTE oversampling to address 20:1 class imbalance
- Executed complex SQL ETL with CTEs and Window functions cleaning raw credit bureau data; engineered 15 predictive features including payment velocity, credit utilization trends, account age
- Applied one-hot encoding for categorical variables and StandardScaler normalization for numerical features prior to model training
- Applied SHAP analysis to isolate Debt-to-Income ratio as primary risk driver, providing interpretable insights for non-technical stakeholders

LEADERSHIP

Housing Office Assistant Sept.2025 – Present
Georgia Institute of Technology Housing Department *Atlanta, GA*

- Coordinate 2+ monthly events serving 200+ students; respond to emergencies (flooding / fire) prioritizing resident safety under high-stress conditions

Chair – DEI Committee, ISyE Department 2023 – 2024
Analytics Working Group Lead, Georgia Institute of Technology

- Analyzed survey data from 500+ respondents; proposed solutions adopted into FY24 strategic plan

TECHNICAL EXPERTISE

Finance & Quant: Bloomberg Terminal API, Time Series Analysis (ARIMA), Monte Carlo Simulation, VaR, Portfolio Optimization
Languages: Python (OOP, Pandas, NumPy), SQL (CTEs, Window Functions), Java, JavaScript, TypeScript, R, Bash
Data Engineering: PostgreSQL, ETL Pipelines, 3NF Modeling, ETL Pipeline design, Data Modelling, EERD
Frameworks & Dev: Django, Flask, React, React Native, Next.js, Docker, Redis, Celery, REST APIs, CI/CD, PyTest, Git
Machine Learning: RAG Pipelines, NLP (Transformers), XGBoost, Scikit-Learn, SHAP, A/B Testing, Experiment Design